

Video Game Programming: Audio and UI

L09

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Role of audio

- Games are not only graphics
- Audio is important to create a fun and engaging game, and can even be an essential part of mechanics
- Audio can contribute to presence and engagement as well as provide feedback to the players
- If audio does not support game theme or outcome, it can distract players and reduce engagement
- It is important to consider formats and other properties of audio resources as well as runtime parameters

Audio types

- Vocalization
 - Non-player characters talking each other or with the player
 - Narrator describing events without taking part in the story
- Music
 - Keeps engaged
 - Arouses emotions, sometimes connected to characters or places
- Sound effects
 - Can be made by game objects (e.g., gun fire)
 - Can provide feedback for UI interaction (e.g., click sound on button press)
- Ambient effects
 - Create atmosphere (e.g., birds, distant sounds)
 - Contribute to presence

Audio in real and game world

- In the real world:

- Audio is emitted by objects and heard by listeners
- Audio perception depends on direction and distance
- Fast-moving audio sources change pitch
- Surroundings affect audio reflection



- In game worlds:

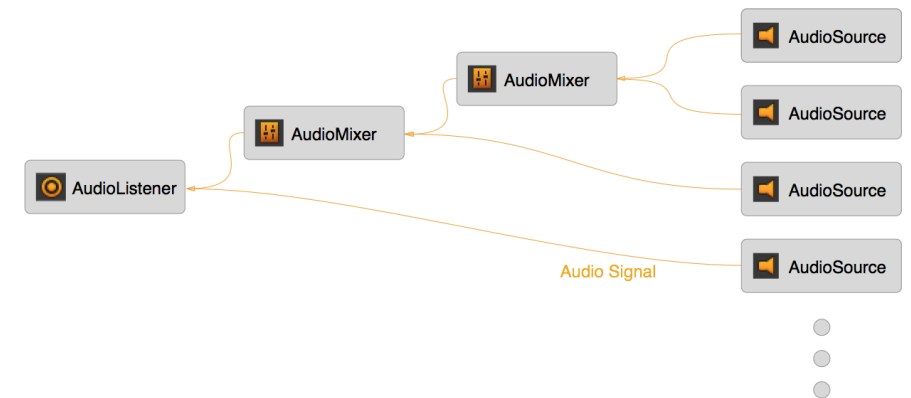
- Audio sources are attached to game object to generate audio from their position
- An audio listener is attached to the main camera to simulate players' ears
- Game engines simulate the effects of source distance and position from listener
- Relative speed of source and listener can be used to simulate Doppler Effect
- Unity can't simulate echoes from scene geometry, but audio filters can be added to sources and reverb zones can be added to the scene (e.g., to simulate a tunnel)

Audio resources

- As for textures, game engines support common audio formats (e.g., wav, mp3), but convert them in more convenient formats for runtime use
- Unity uses PCM, ADPCM or compressed formats (varying with platform)
- Compressed audio uses less space on disk, but takes time to decompress
- Compressed audio is useful for huge music files, while uncompressed audio is useful for sound effects to be played without noticeable delay
- Different compression formats (codecs) can be lossy or lossless
- Audio files can be mono (1 channel) or stereo (2 channels)
- Unity also supports tracker modules (evolution of MIDI, e.g., xm, mod), which can allow to reduce size without compromising quality
- Sound and music can be found online, but reuse depends on copyright

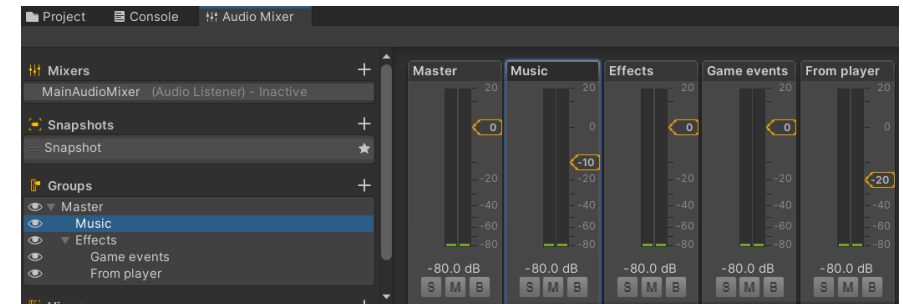
Audio sources

- Audio resources are played from audio sources attached to objects
- Sources play clips as 2D or 3D audio (without or with position)
- Sources have a priority, a volume and a max distance after which they are not perceived
- Volume can roll off following predefined or custom curves
- Audio is sent directly to the listener or passes through mixers (in fig.)



Audio mixer

- Audio balancing can be difficult (e.g., music can cover other sounds, ambient may sound louder than an explosion,...)
- Unity AudioMixers allows organizing sounds into a tree of groups (e.g., for music, sound effects, ambient audio,...)
- Unity offers a mixing desk to adjust group volumes and apply effects (in fig.)
- Mixer states can be saved in snapshots, allowing for different types of transitions during gameplay



Audio filters and effects

- Audio filter can be used to apply effects to listener or sources
- In addition, a variety of audio effects can be applied to audio mixer groups
- A very useful effect is Duck Volume:
 - It takes input from an audio group and drops the volume of another
 - It is used to drop music volume whenever dialogs are played
 - It is also useful also to drop all audio when playing a notification sound
 - A Duck Volume effect should be added to the group to mute
 - A Send effect should be added to the group that causes the other to mute

Audio scripting

- Audio file properties such as length, channel count and sample rate are available from scripts (in Unity as Audio Clip instances) even before the audio data is loaded
- Scripts can also control audio sources to play different audio clips and change real-time properties (e.g., volume)
- To change AudioManager parameters, such as those of effects or audio levels, those properties must be exposed

(Graphical) User Interface

- A User Interface (UI) enables interaction between humans and a machine or computer
- A graphical user interface (GUI) is a UI based on visual metaphors
- The term GUI is used to distinguish GUIs from text-based interfaces (CLIs)
- Games can have both GUIs and other kinds of UIs (e.g., based on physical movements)
- UIs in games typically include camera or main character control and a GUI for the main menu



Angry Birds



Rainbow six siege

UI design

- Determine UI functionality required for a project
 - What choices can players make (input)?
 - How can the effect of a choice be made clear (output)?
 - Which UI elements are available to make choices (e.g., buttons, menus)?
- Understand how UI affects usability and user experience
 - Are UI controls easy to learn and use?
 - How players can be encouraged to correctly interact with UI elements?
- Find a cool and appropriate graphical theme
 - How do UIs of current games for the same genre / platform look like?

Diegetic and non-diegetic UIs

- Diegetic UIs
 - The UI is fully integrated with the game world (usually attached to a game object)
 - E.g., a menu embedded in a device (in fig.)
- Non-diegetic UIs
 - The UI is rendered outside the game world (usually an overlay to the camera)
 - E.g., a HUD in a third-person game (in fig.)



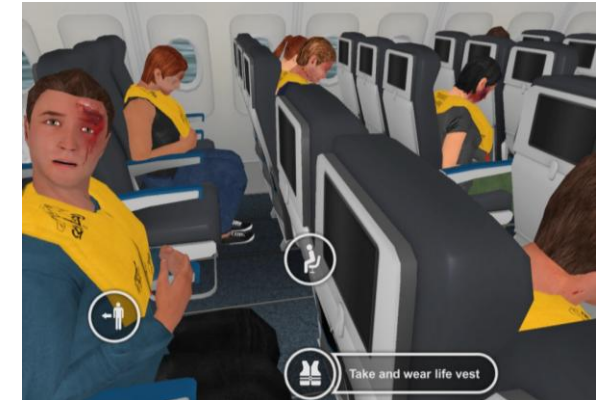
Dead Space diegetic UI



Horizon Zero Dawn non-diegetic UI

UIs in different platforms

- UIs significantly vary in different platforms
- Different platforms support different interaction devices (e.g., mouse, gamepad, touch, 6-DOF tracked controllers)
- Some platforms need responsive UIs
- Some UI choices may be optimal for a platform and unsuitable for another
- Different genres require different types of input and output (e.g., menus, health bars, object interactions) thus affecting UI



Prepare for Impact UI
for VR and mobile

UI design exercises

- Think about what you like and dislike in UIs of the games you play
- Focus on one game platform, genre, and concept
- Create several sketches for the UI of chosen game concept
- Think about what controls will be needed
- Find a design for UI elements you would like to include

Different methods for UI development

- Game engines support different UI implementations
- Unity supports:
 - A UI toolkit based on standard web formats
 - A Unity UI based on game objects
 - An immediate mode UI based only on script code
- Unity UI development consists in a hierarchy of UI game objects (e.g., panels, labels, buttons) attached to a canvas
- Canvas can be rendered both in-game and as an overlay
- Unity UI has also options to implement responsive UIs

Bibliografy

- Coggan, Unity Game Audio Implementation: A Practical Guide for Beginners, Focal Press, 2021
- Gomila Salas Juan Gabriel, Borromeo Nicolas Alejandro, Hands-On Unity Game Development, 4th ed., Packt Publishing, 2024
- Murray, C# Game Programming Cookbook for Unity 3D, 2nd ed., CRC Press, 2021



UNITY GAME AUDIO IMPLEMENTATION A PRACTICAL GUIDE FOR BEGINNERS

ANDREW COGGAN

A Focal Press Book

